Ovarian and Mullerian Infertility

Jackie Gutmann, M.D.
RMA at Jefferson
jackieg@ivirma.com





Disclosure

- I have nothing to disclose
- I did not use AI to prepare this lecture

Female Infertility

- **Learning Objectives:**
  - At the conclusion of this session, the participant should be able to:
    - Describe the impact of aging on (ovarian) fertility.
    - Define infertility.
    - List and describe the causes of (ovarian/mullerian) infertility.
    - Describe the evaluation of infertility.
    - Identify treatment options available.

Study Question 1

A 32 year old cis-female nulligravida in a heterosexual relationship presents for evaluation. The couple has not used contraception for the last 15 months. She has regular menses occurring every 30-32 days. She has a history of chlamydia as a teenager which required hospitalization. Her partner is 39 and had not previously contributed to a pregnancy. Her examination is unremarkable.

a. What is the most likely cause of this couple's infertility?
b. What are the next steps in the evaluation of this couple?
c. What treatment recommendations would you make?

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Study Question 2

A 27 year old nulligravida cis-female in a same sex marriage presents for evaluation. She was placed on oral contraceptives (OCP) at the age of 15 for cycle control. She discontinued the OCP approximately 9 months ago. She has had 2 menses since that time. Her BMI is 23.9. The remainder of the examination is remarkable for mild hirsutism. The couple is interested in achieving a pregnancy.

a. What barriers exist to achieving pregnancy in this couple?
b. What are the next steps in the evaluation of this couple?
c. What treatment recommendations would you make?

Study Question 3

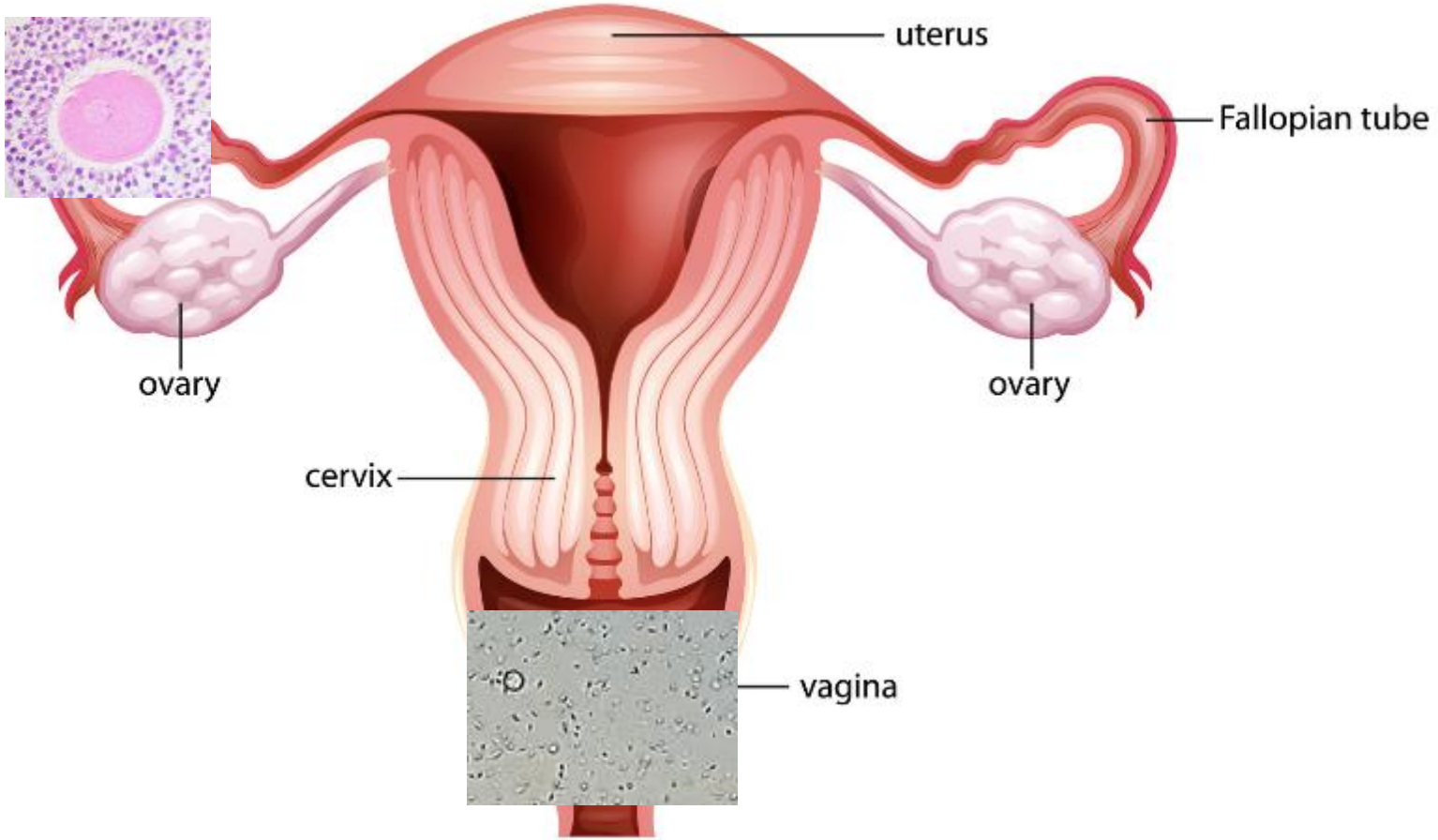
A 44-year-old cis-female with 2 children ages 12 and 14 from a prior relationship is now remarried and presents for evaluation. They have not used contraception for the last 6 months; she is confused and disappointed as her prior pregnancies were conceived easily. She has regular menses occurring every 26-28 days. She has no history suggestive of pelvic surgery or infection. Her partner is 39 and has 2 children from a prior relationship. Her examination is unremarkable.

a. What is the most likely cause of this couple's infertility?

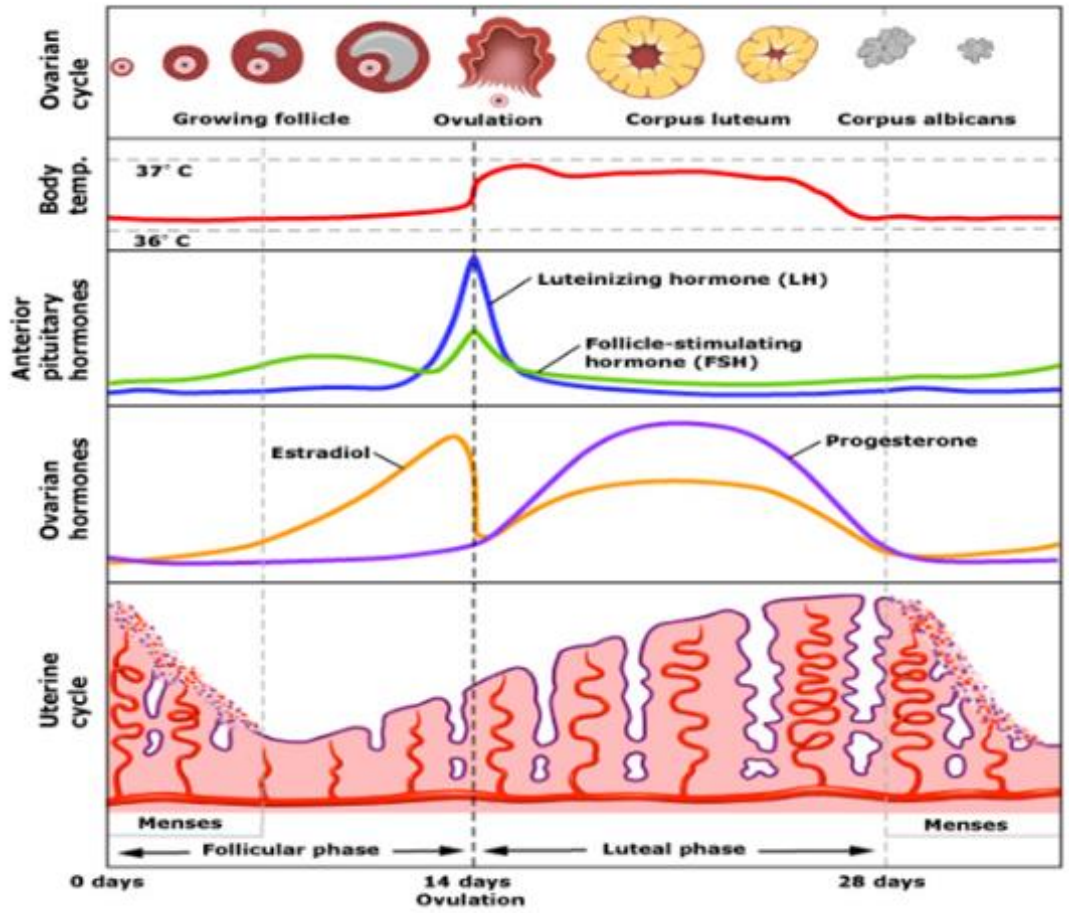
b. What are the next steps in the evaluation of this couple?

c. What treatment recommendations would you make?

Ingredients for Fertility

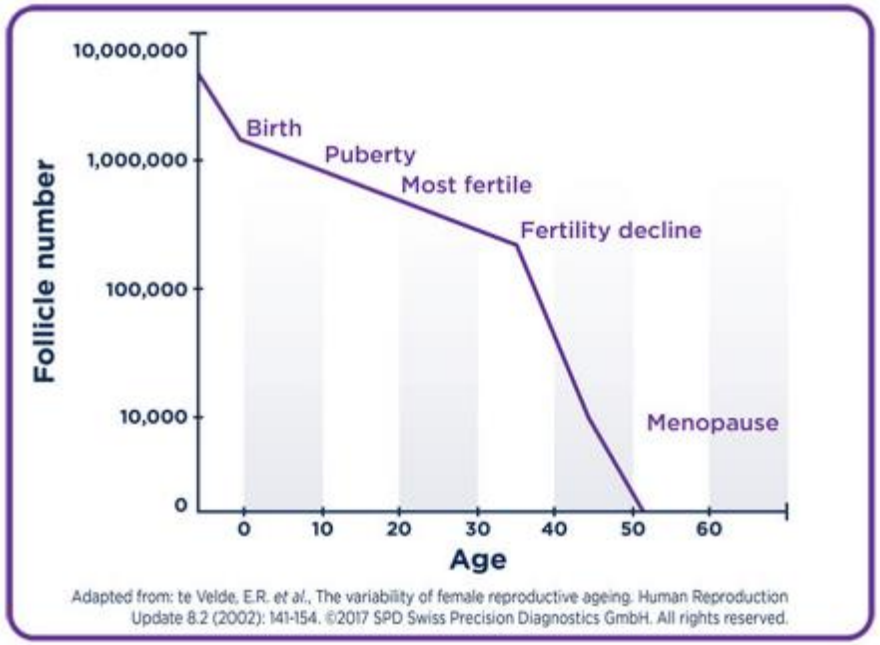


Menstrual Cycle

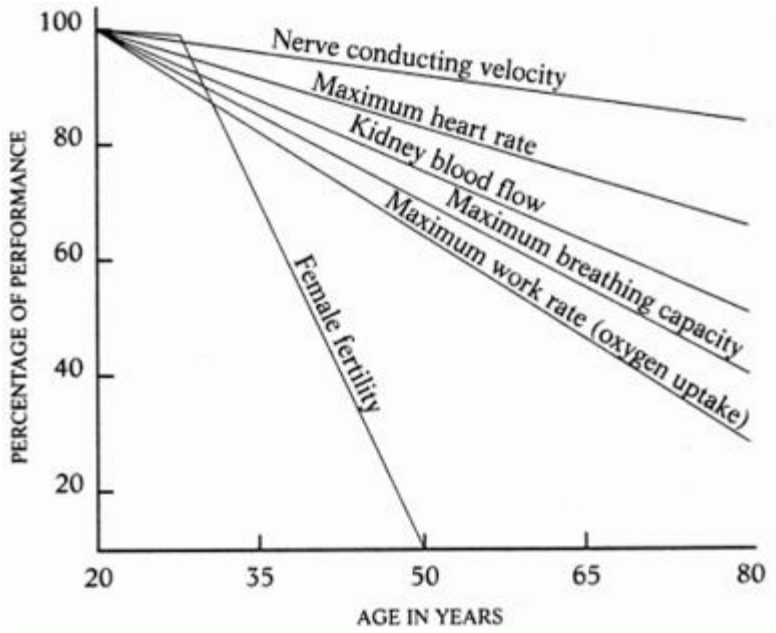


Reproductive Aging



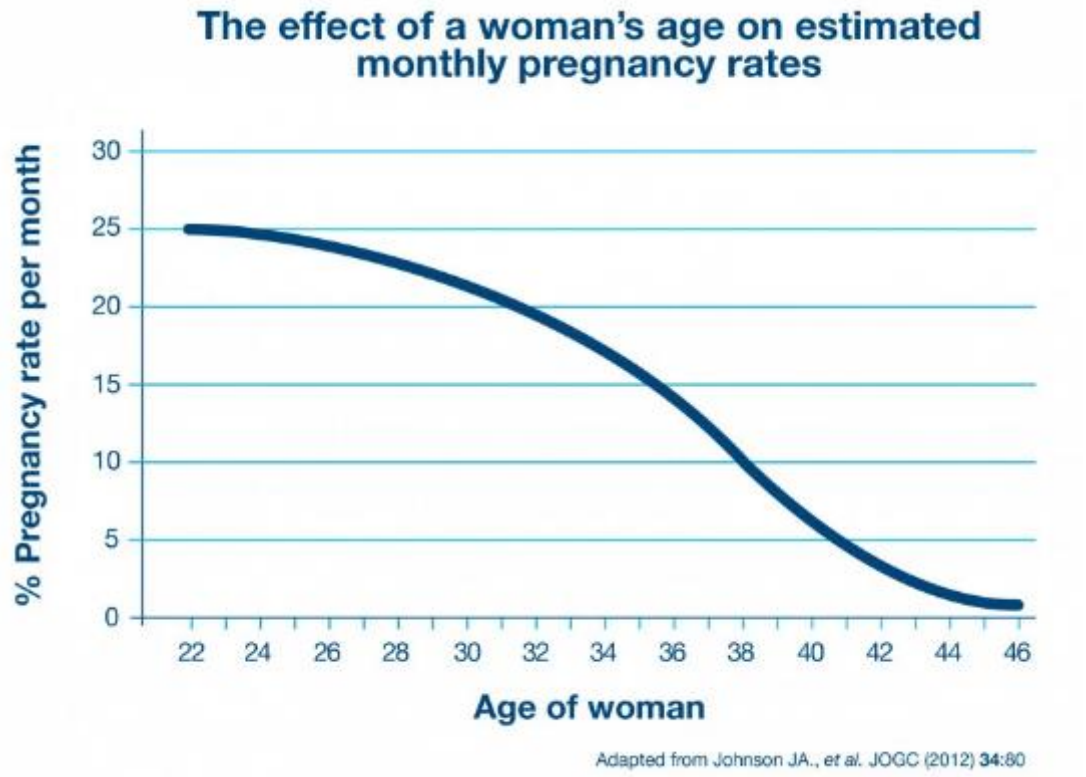


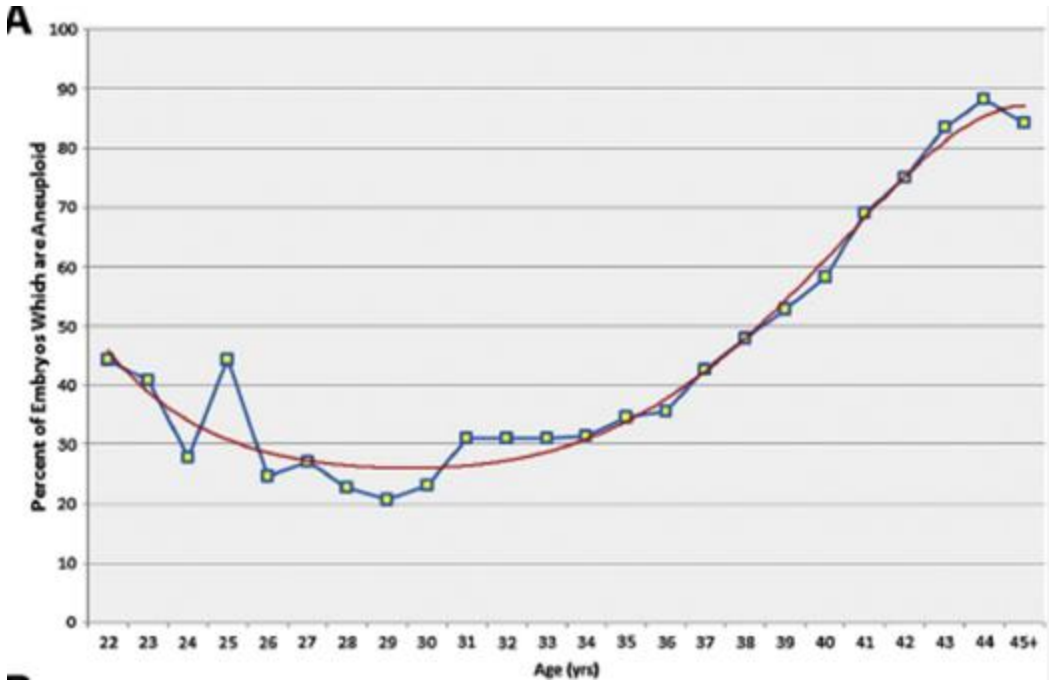
Adapted from: te Velde, E.R. *et al.*, The variability of female reproductive ageing. Human Reproduction Update 8.2 (2002): 141-154. ©2017 SPD Swiss Precision Diagnostics GmbH. All rights reserved.



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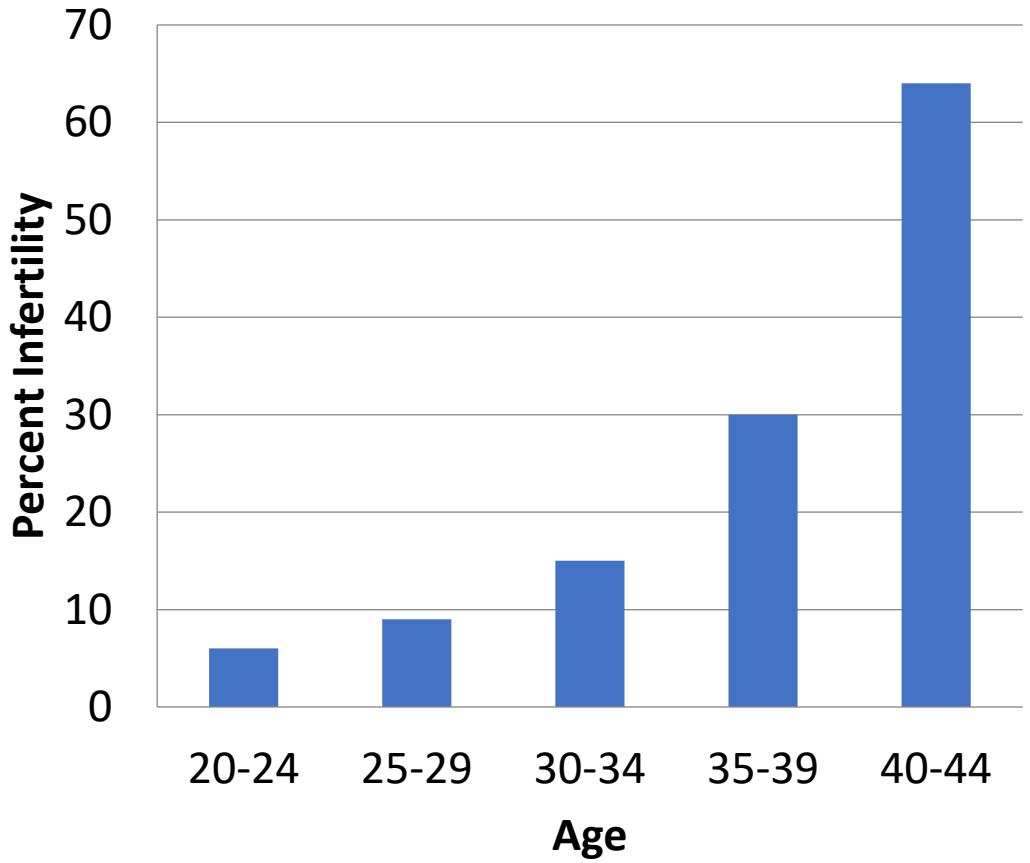
Reproductive Aging





Franasiak, Fertil Steril, 2015

Aging and (In)Fertility

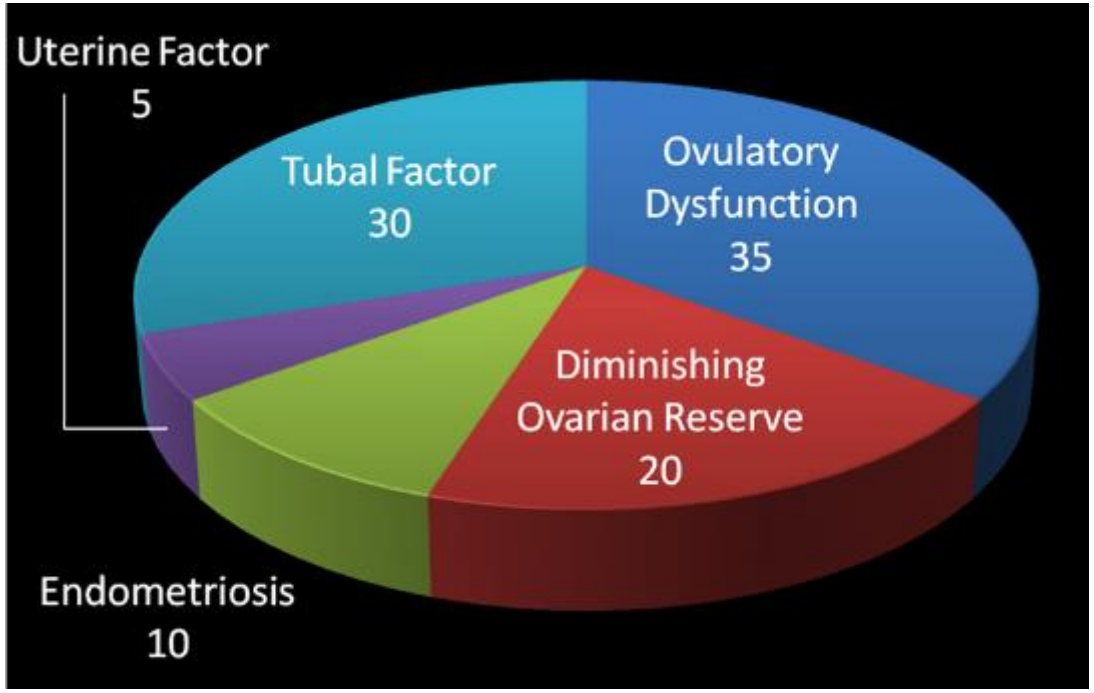


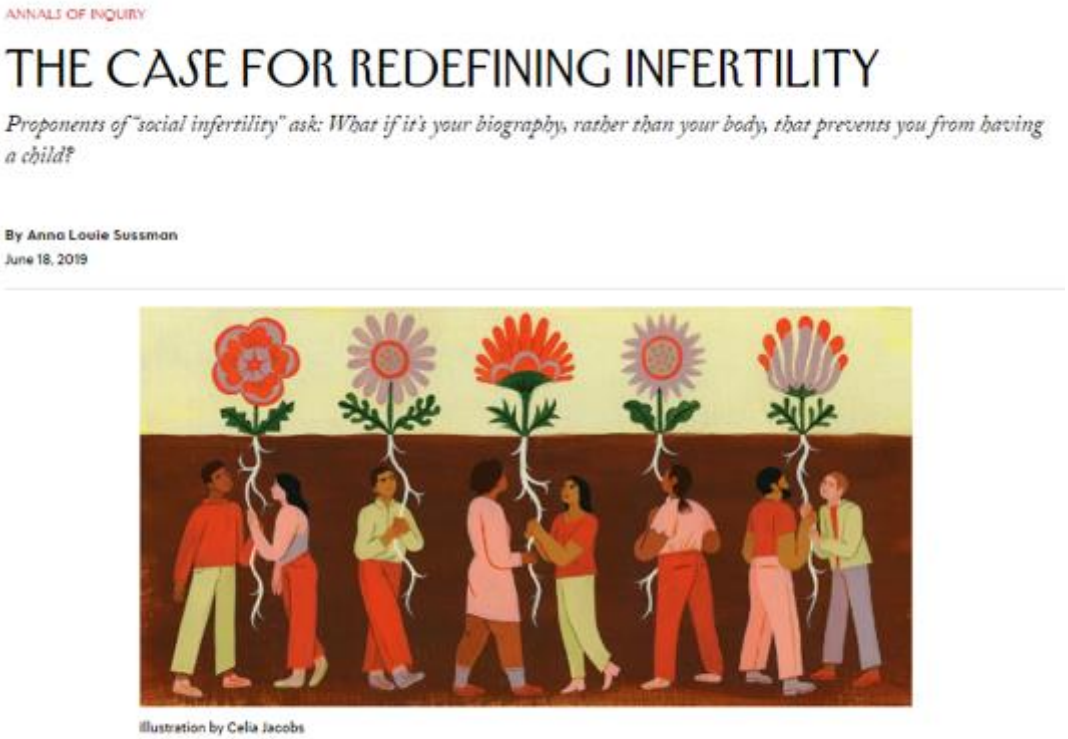
(What is) Infertility:

- Inability to conceive despite attempting for one year
  - If age > 35 or previous pregnancy, this interval shortens to 6 months

Menkin, 1986

Causes of Infertility





ANNALS OF INQUIRY

THE CASE FOR REDEFINING INFERTILITY

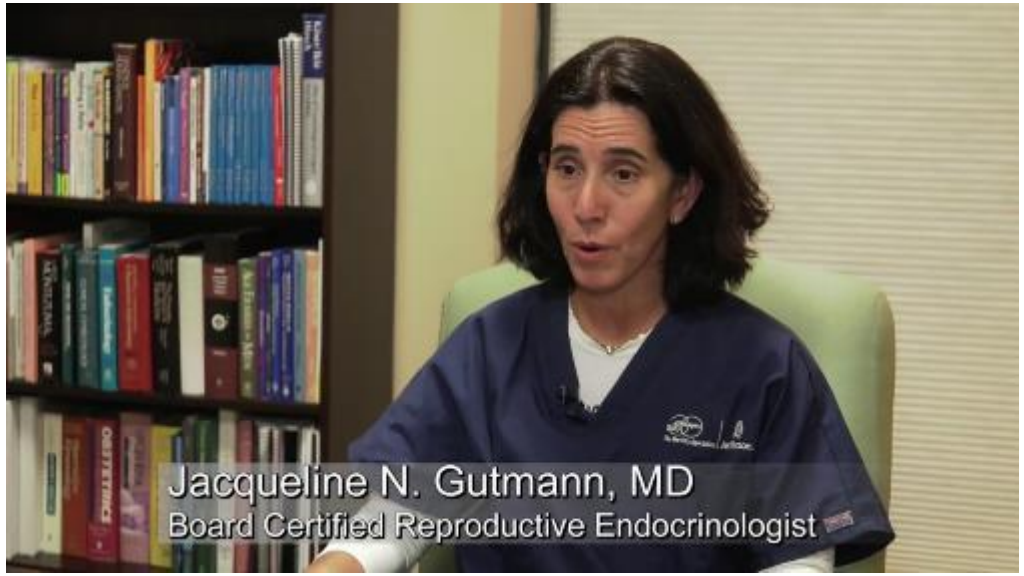
Proponents of "social infertility" ask: What if it's your biography, rather than your body, that prevents you from having a child?

By Anna Louie Sussman
June 18, 2019

Illustration by Celia Jacobs

Evaluation of Infertility

- History
- Physical exam
- Testing



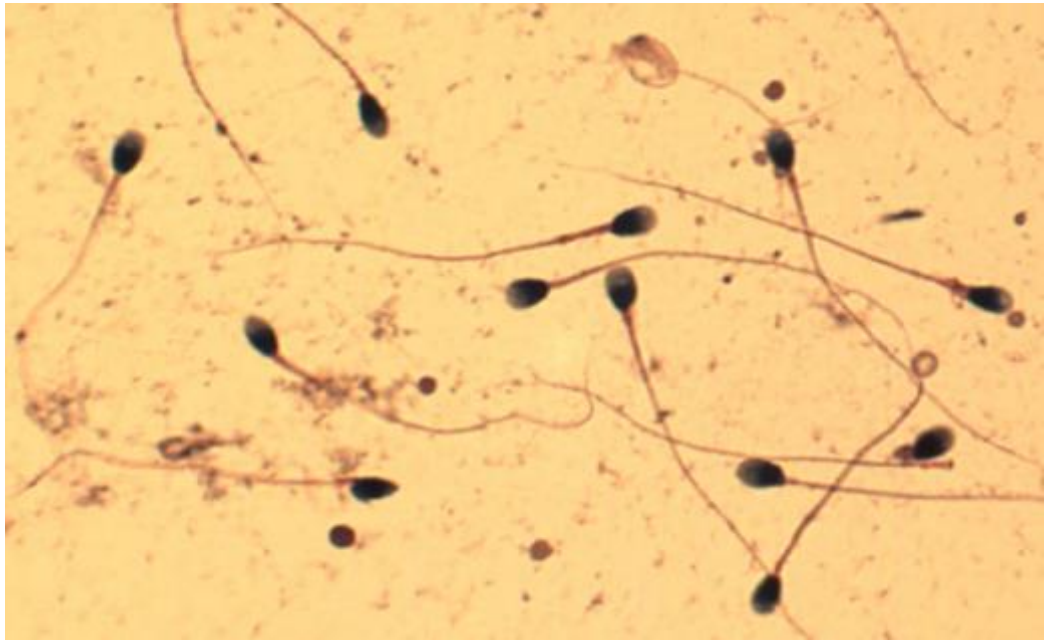
Sperm Related Fertility: Evaluation

- Instructions
  - Abstinence, 2-3 days
  - Sterile container (preferable)
  - Specimen to lab within 2 hours
  - Specimen kept at body temperature (protected from cold)



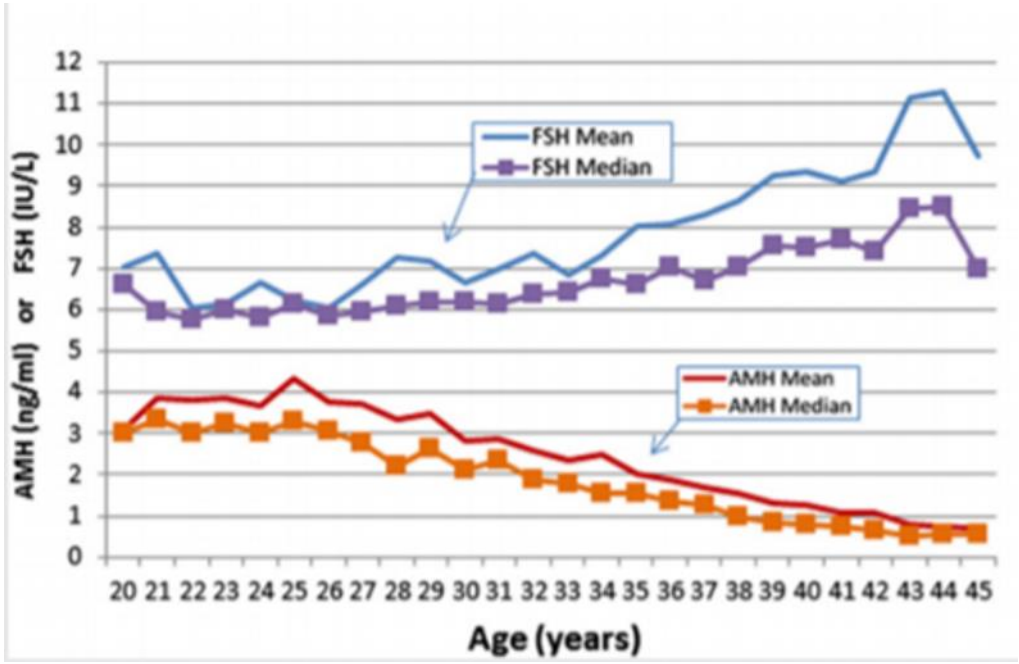
Sperm Related Fertility: Evaluation

- Semen analysis
  - Volume -1.4ml
  - Count -16 mil/ml
  - Motility - 42%
  - Morphology - 4%
- Exam (urologist)
- Blood work
- Other sperm tests

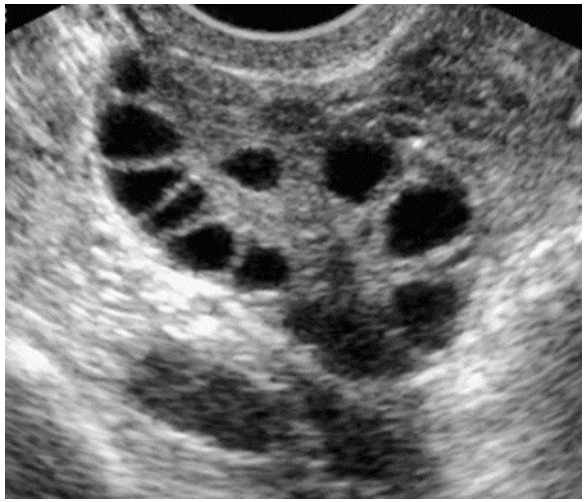


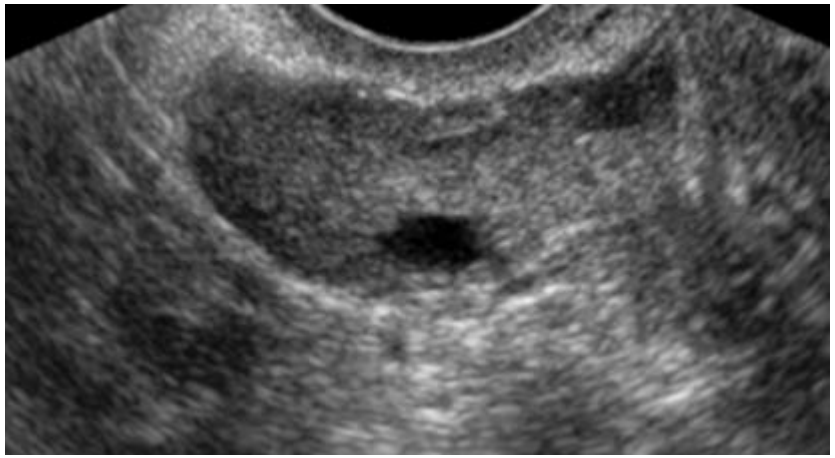
Ovarian Reserve Assessment

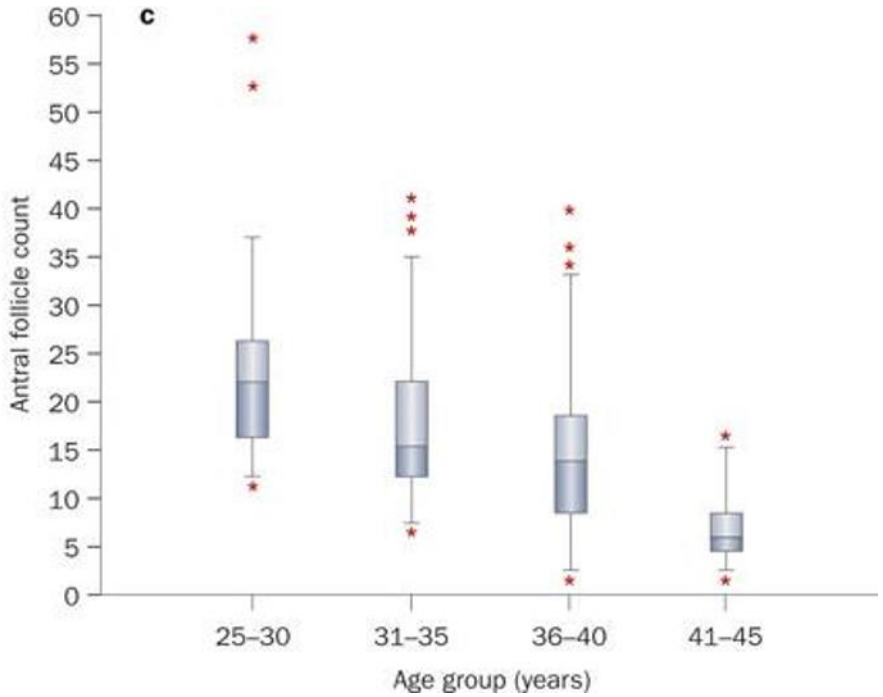
- Ovarian Reserve Testing
  - Anti-Mullerian Hormone level(AMH)
  - Baseline Follicle Stimulating Hormone (FSH)
  - Basal Antral Follicle Count (AFC)



Ovarian Reserve Assessment

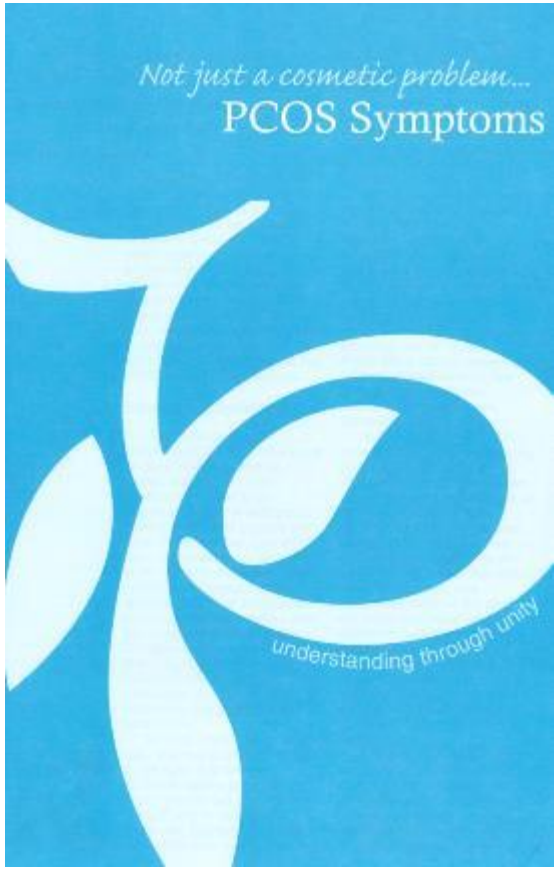






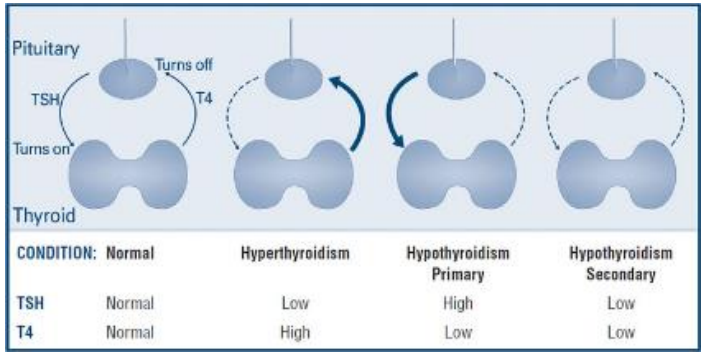
Ovulatory Dysfunction

- Polyendocrine Metabolic Ovarian Syndrome (PMOS)
  - Polycystic ovarian syndrome (PCOS)
- Thyroid disease
- Hyperprolactinemia
- Adrenal gland disorders
  - Late onset congenital adrenal hyperplasia
  - Hyperplasia
  - Cushing's Syndrome
- Other



Ovulatory Dysfunction: Evaluation

- Thyroid hormones
- Prolactin level
- Androgens
  - Testosterone
  - Adrenal hormones
    - DHEAS
    - 17OHP

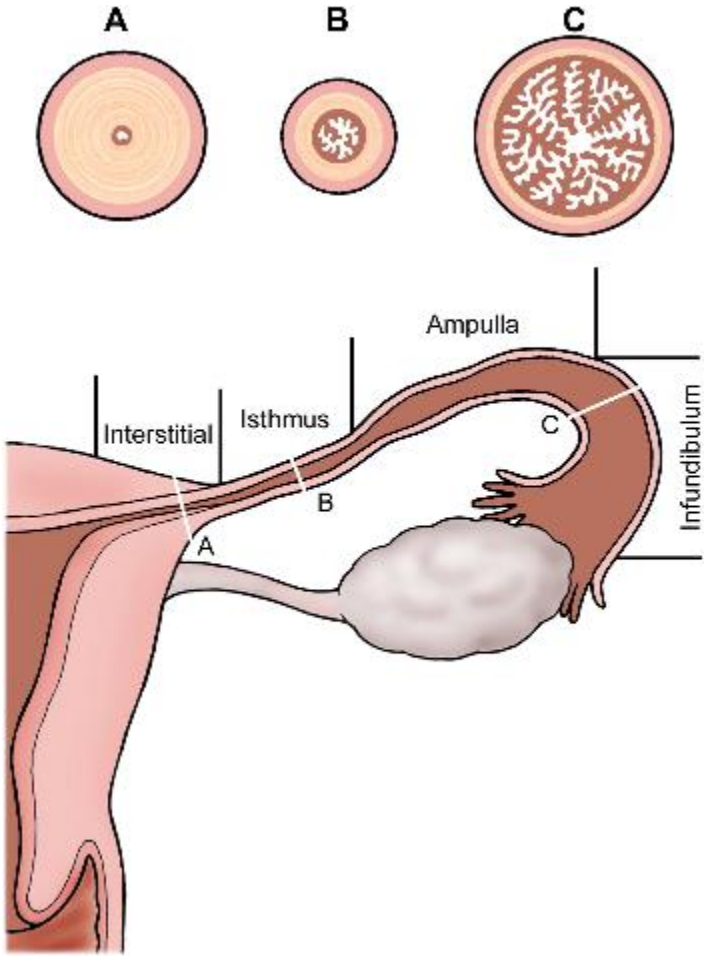


CONDITION:	Normal	Hyperthyroidism	Hypothyroidism Primary	Hypothyroidism Secondary
TSH	Normal	Low	High	Low
T4	Normal	High	Low	Low

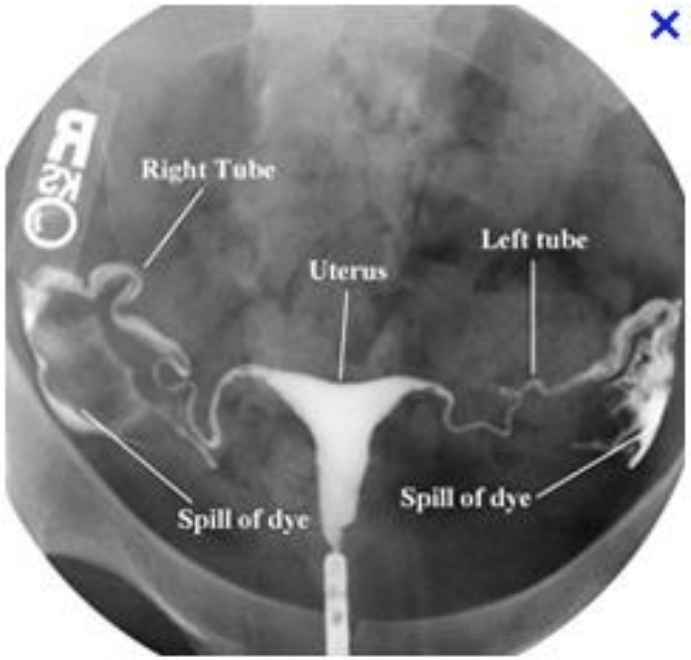


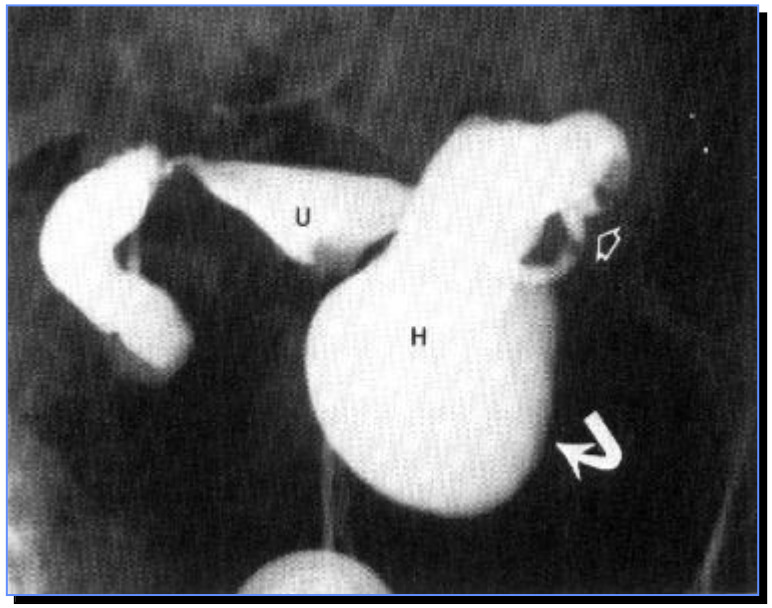
Tubal Disease

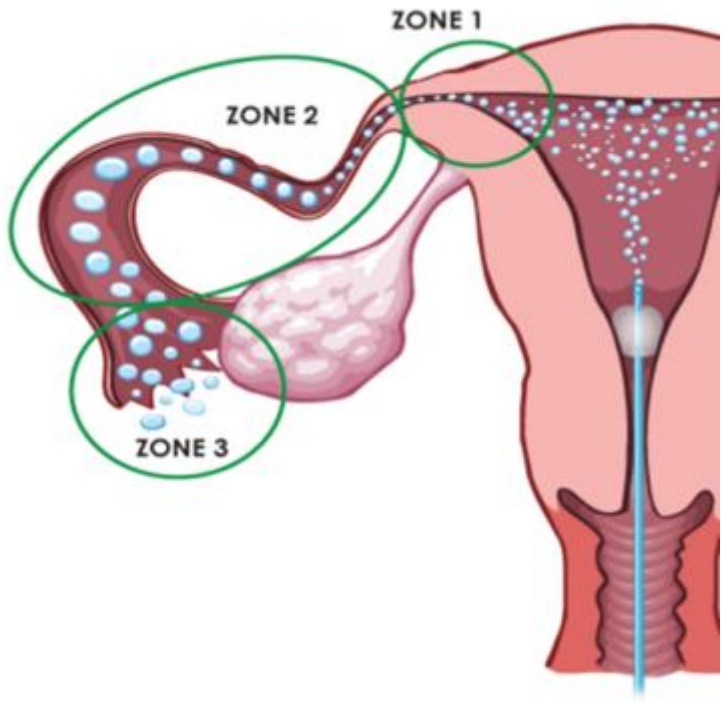
- Proximal occlusion
- Distal occlusion
  - Hydrosalpinx
- Pelvic Adhesions



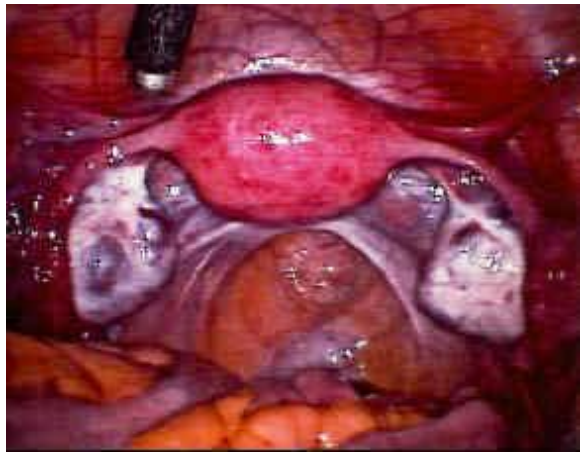
Tubal Disease: Evaluation

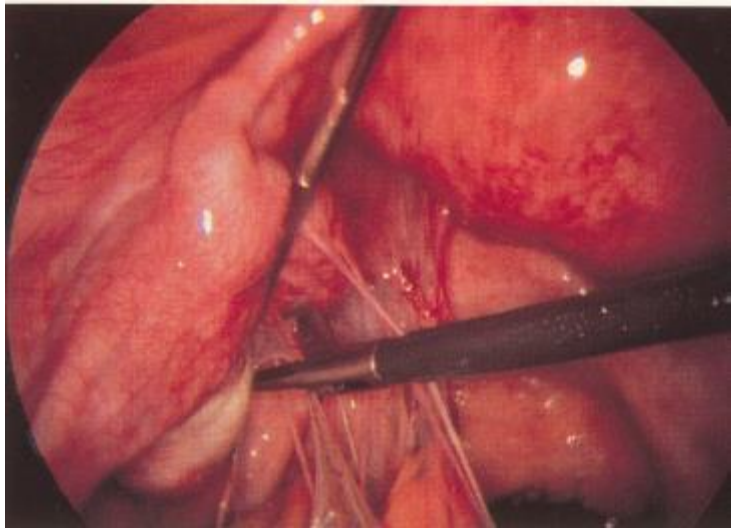


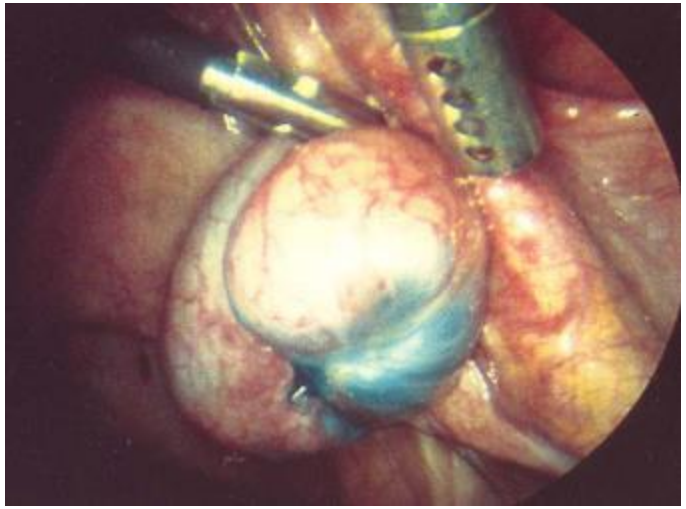




Tubal Disease: Evaluation

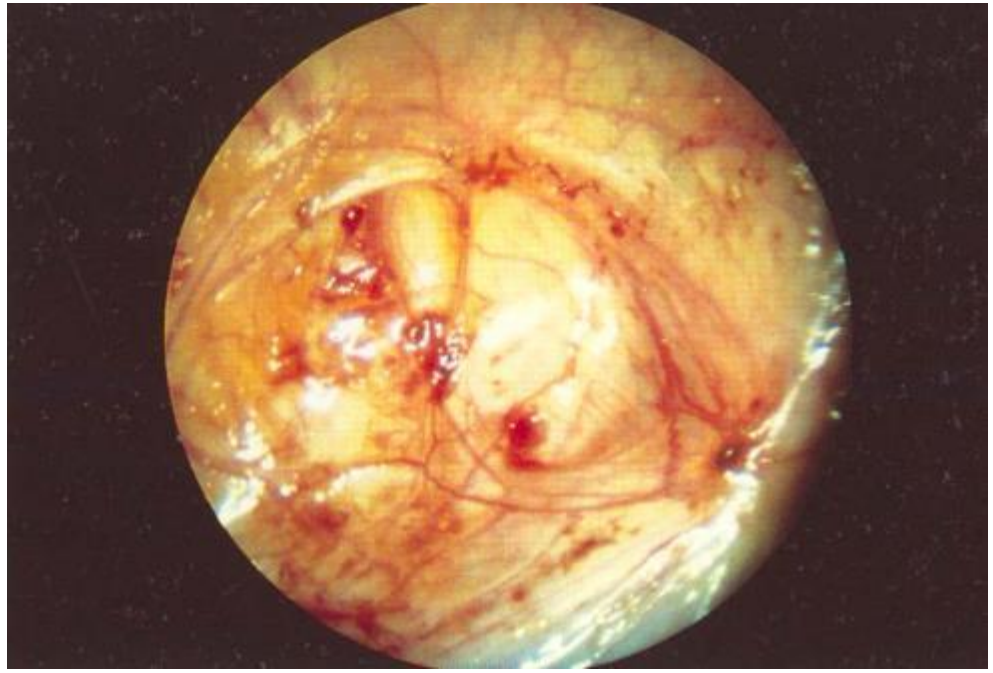






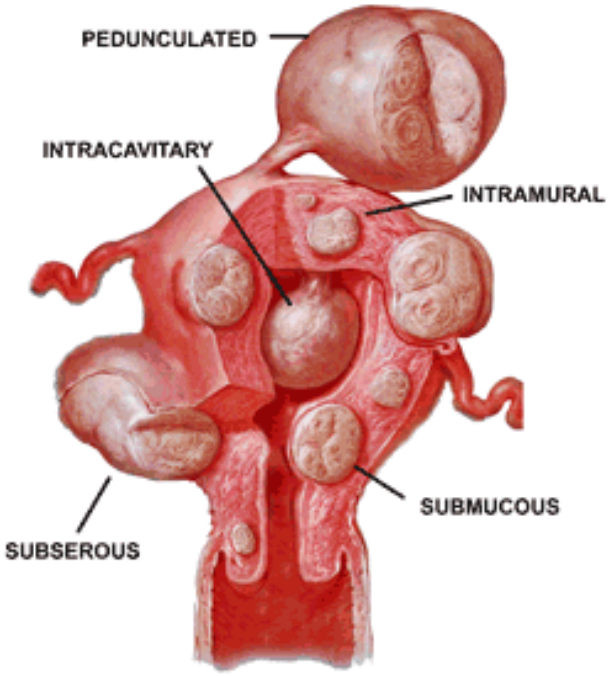
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Endometriosis: Evaluation

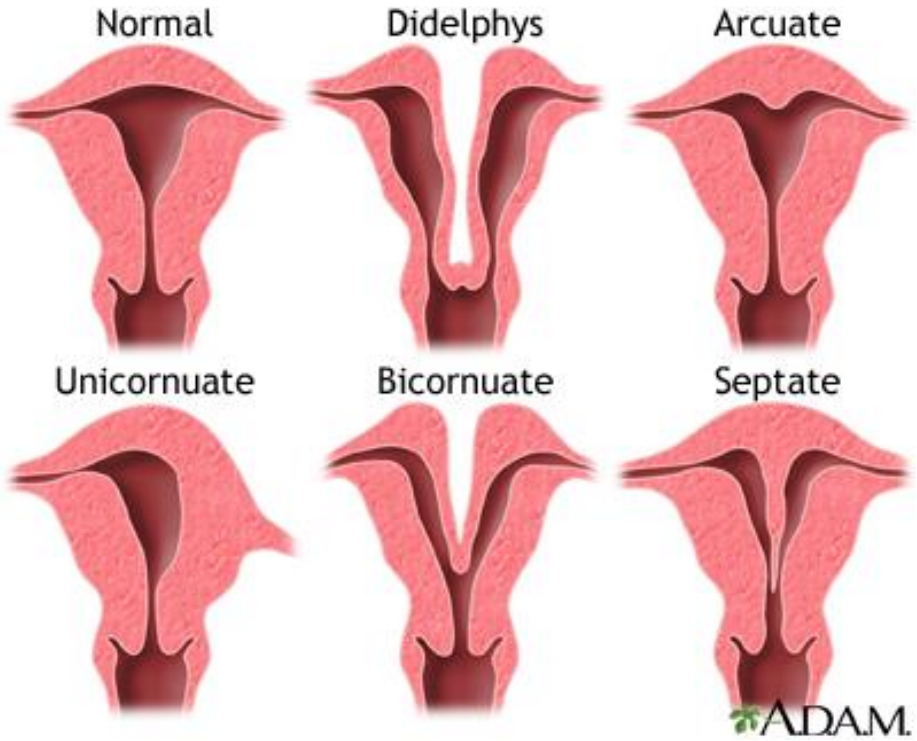




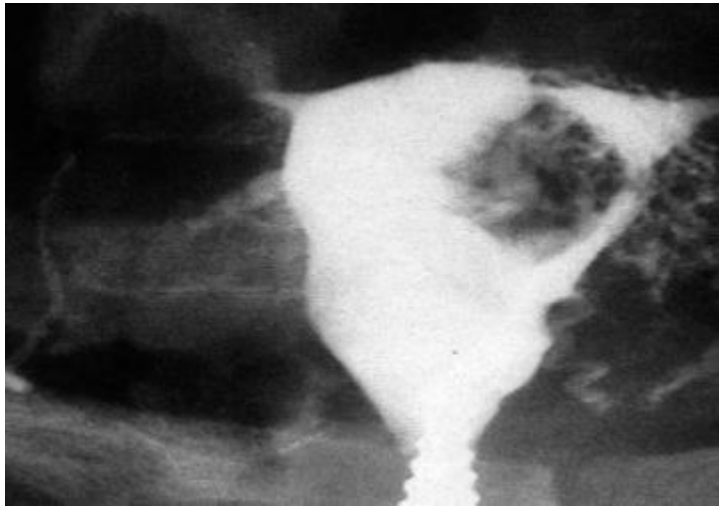
Uterine Abnormalities



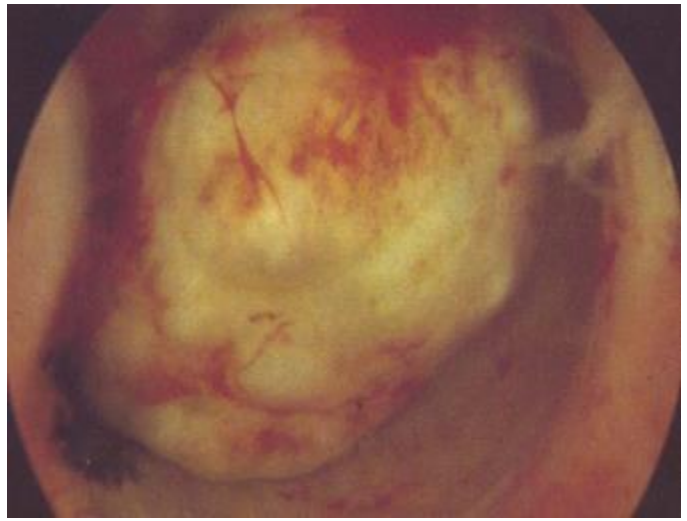




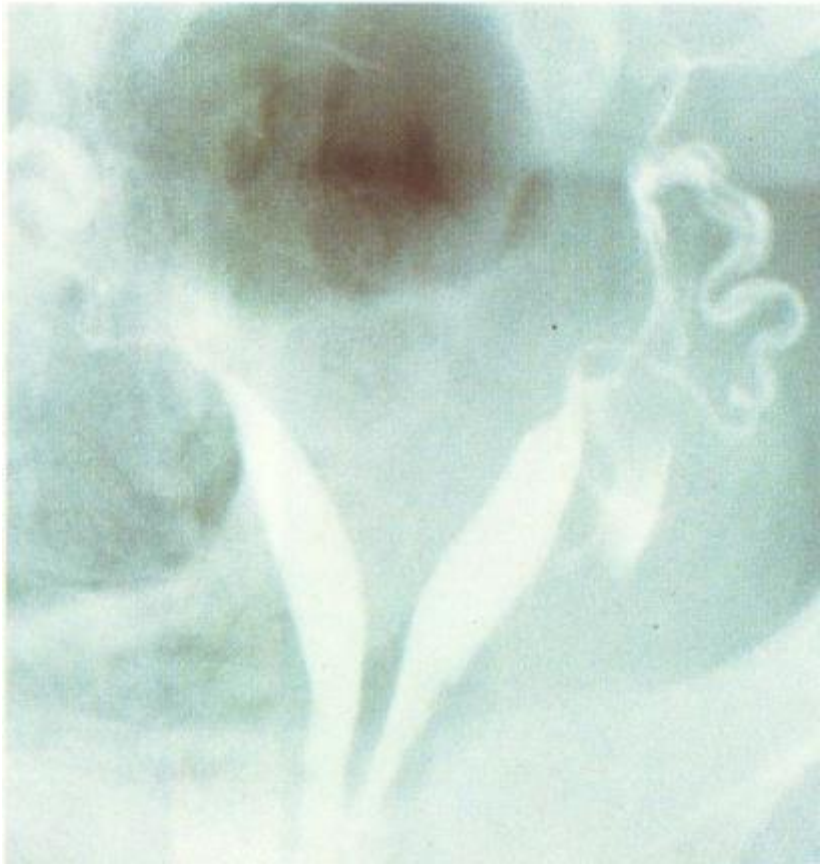
Uterine Abnormalities: Evaluation







Uterine Abnormalities: Evaluation

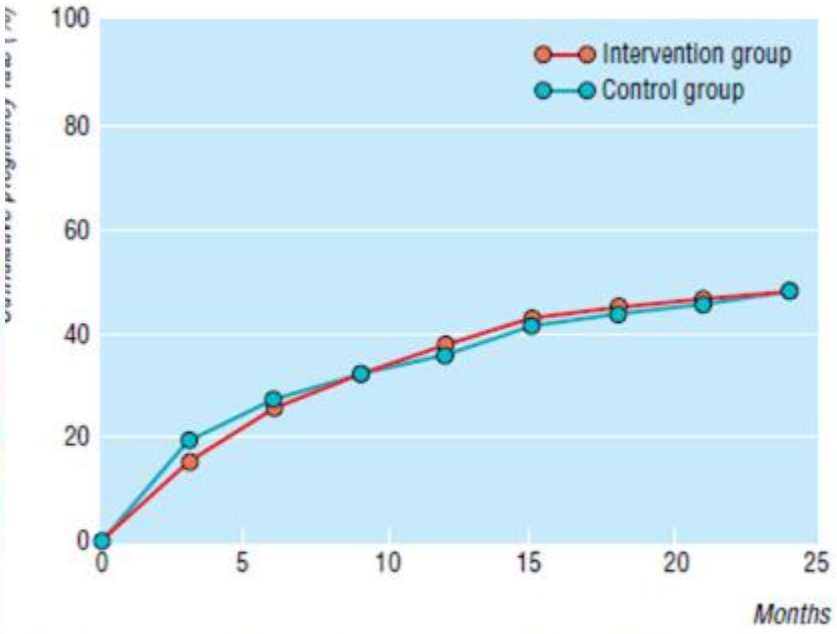




Cervical Mucus Abnormalities: Evaluation







umulative pregnancy rates for 227 couples in intervention group
vhich included postcoital test) and 217 couples in control group
vhich excluded the test)

Unexplained Infertility

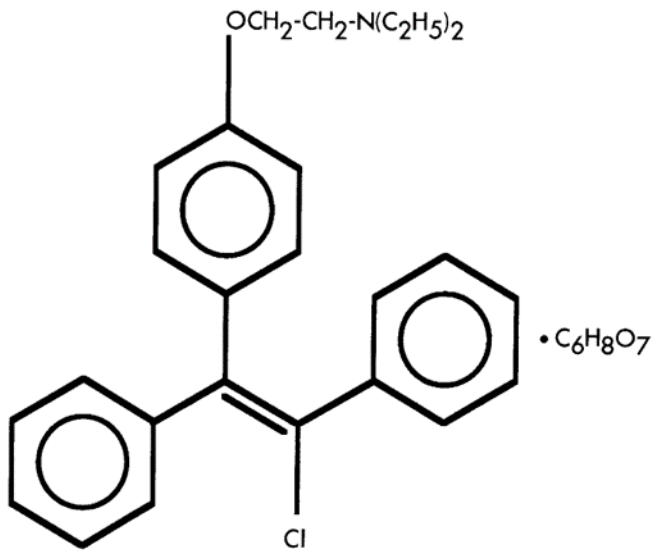


Ovulatory Dysfunction: Treatment

- Treat the underlying abnormality



Ovulatory Dysfunction: Treatment

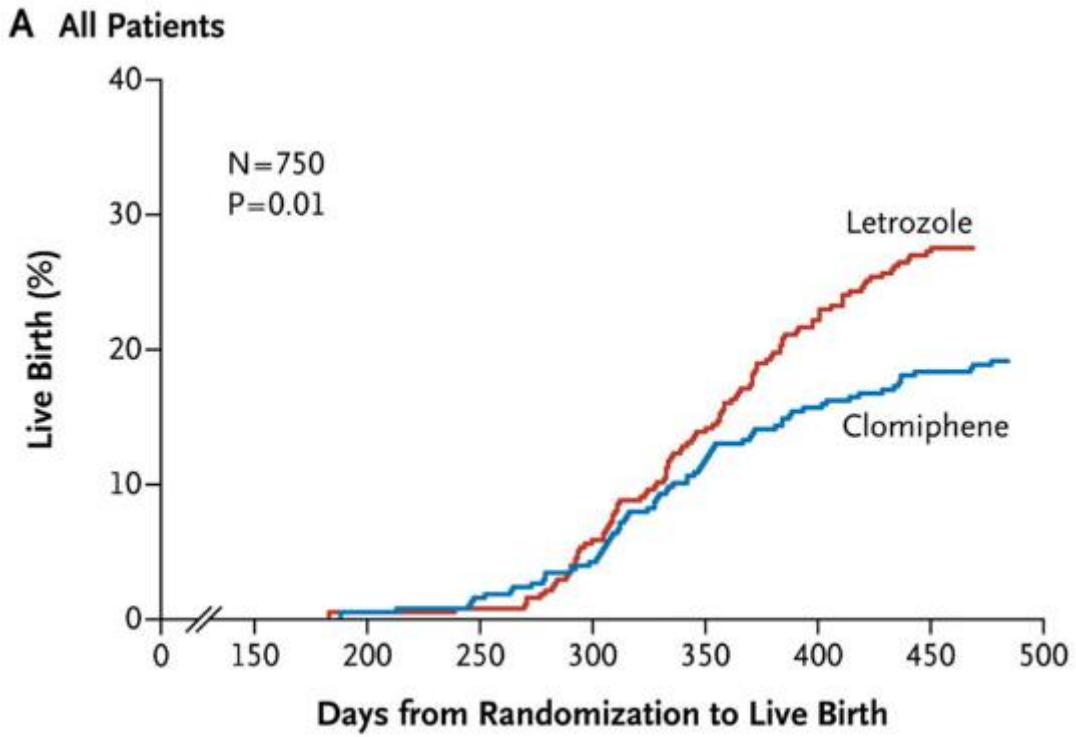


2-[p-(2-chloro-1,2-diphenylvinyl) phenoxy] triethylamine dihydrogen citrate

Fig. 32-1. Structural formula of clomiphene citrate. (Redrawn from Adashi EY: *Semin Reprod Endocrinol* 4:255, 1980.)



Ovulatory Dysfunction: Treatment

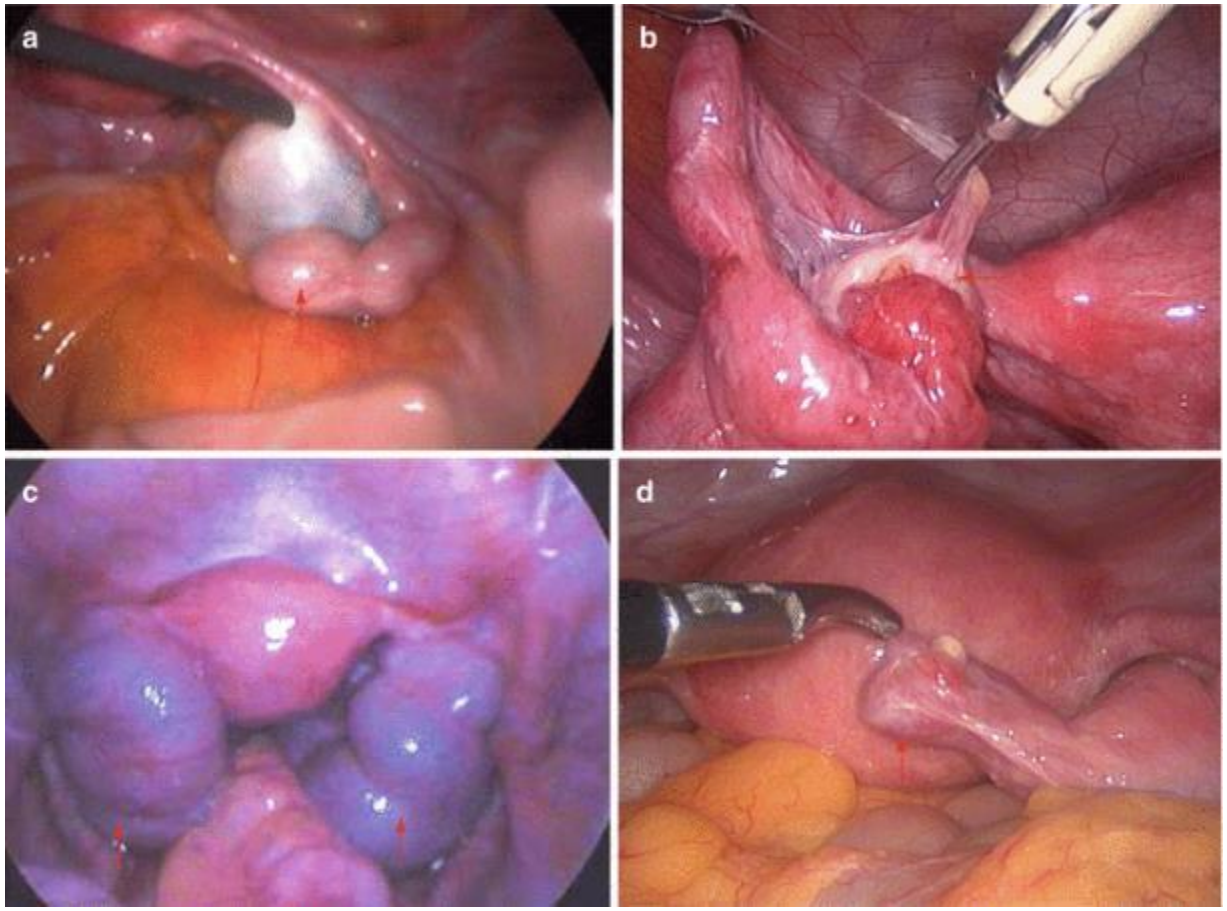


Legro, NEJM, 2014

Ovulatory Dysfunction: Treatment

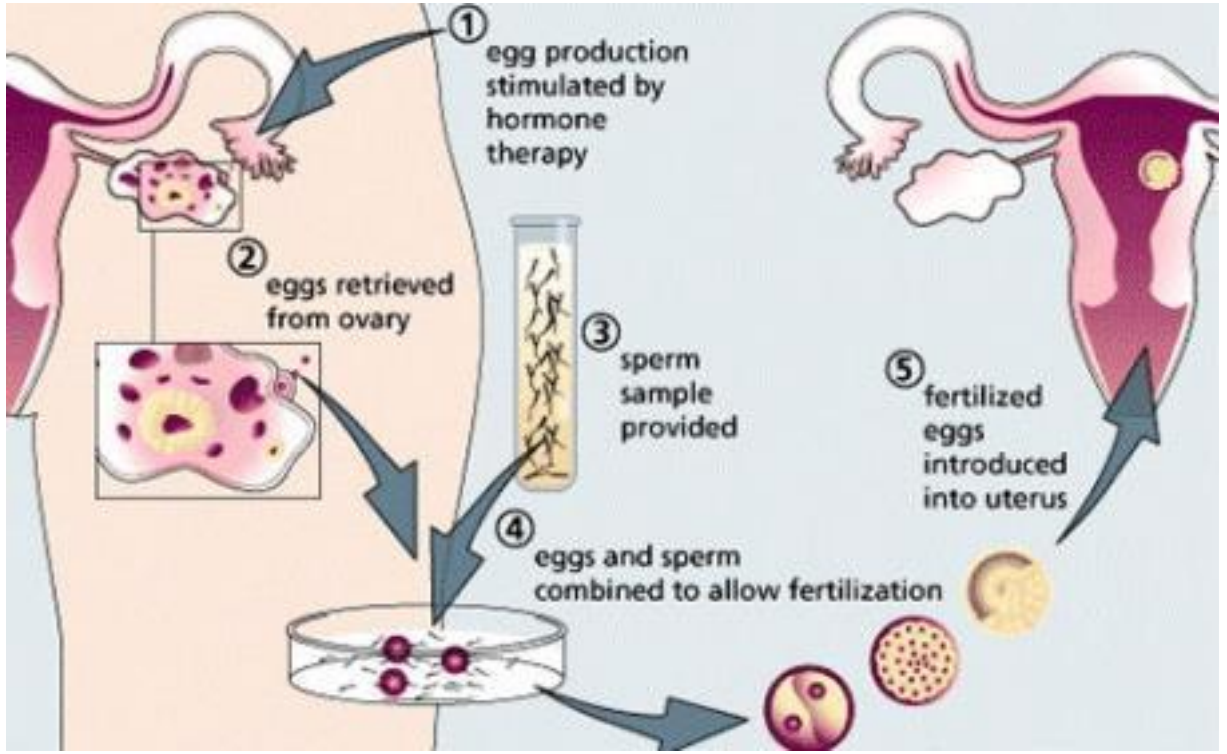


Tubal Disease: Treatment

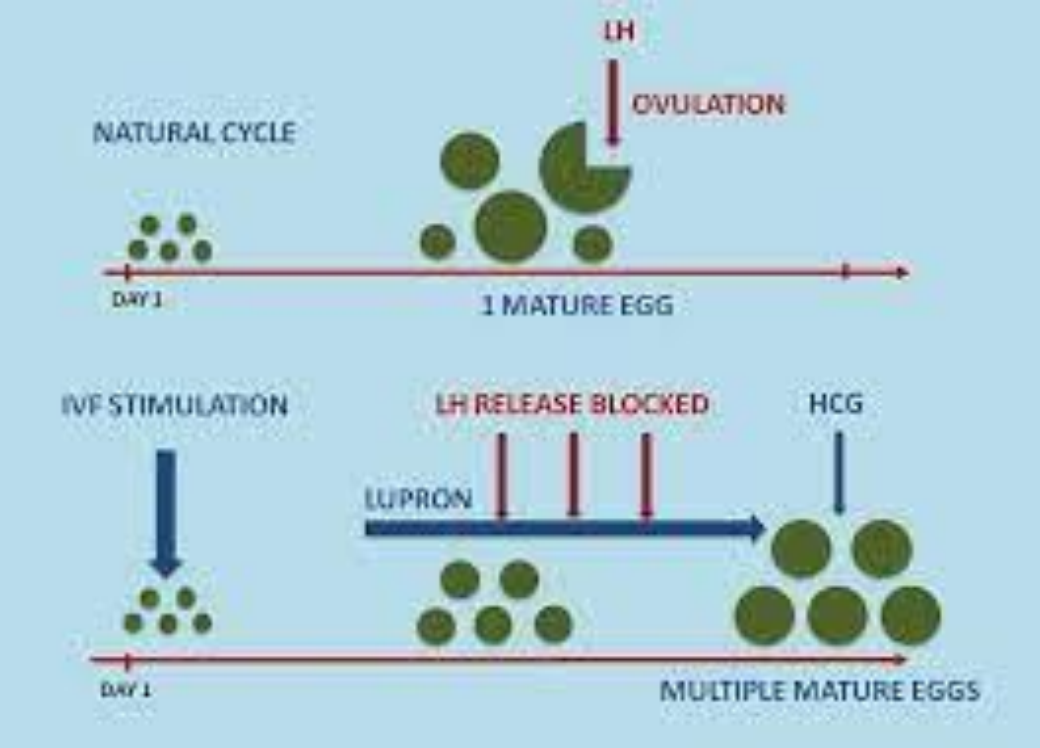




Tubal Disease: Treatment



IVF: Ovarian Stimulation



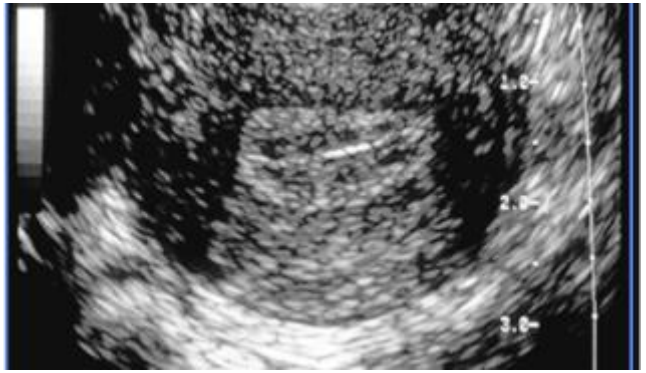


Ovarian Stimulation: Monitoring









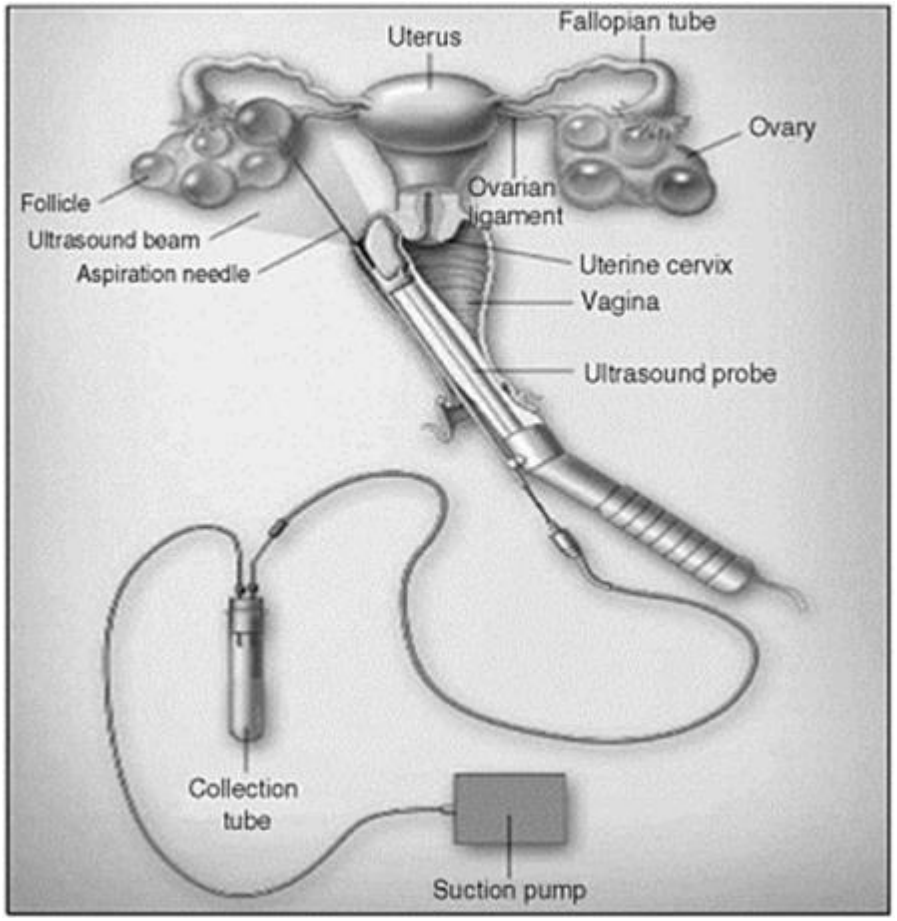
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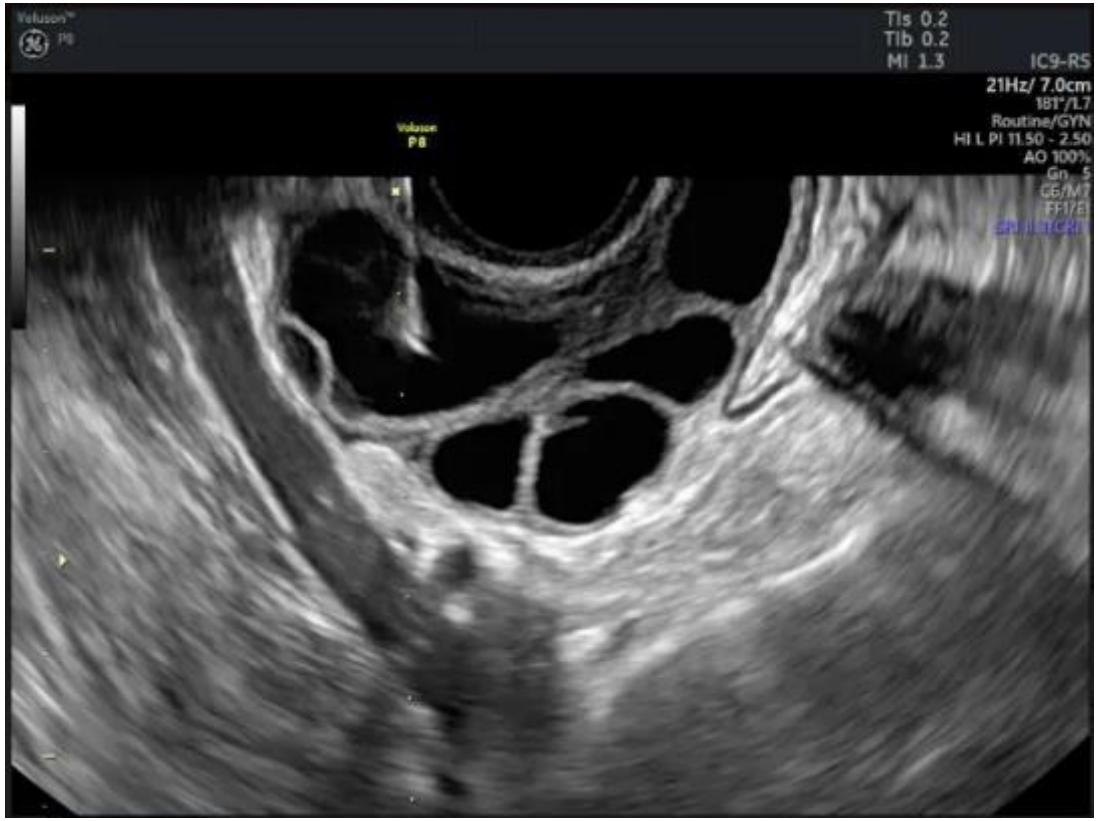
Oocyte Retrieval

- Outpatient procedure
- IV sedation
- Duration of procedure
  - 5-20 minutes
- Recovery
  - 30-45 minutes
- Risks
  - Bleeding, infection



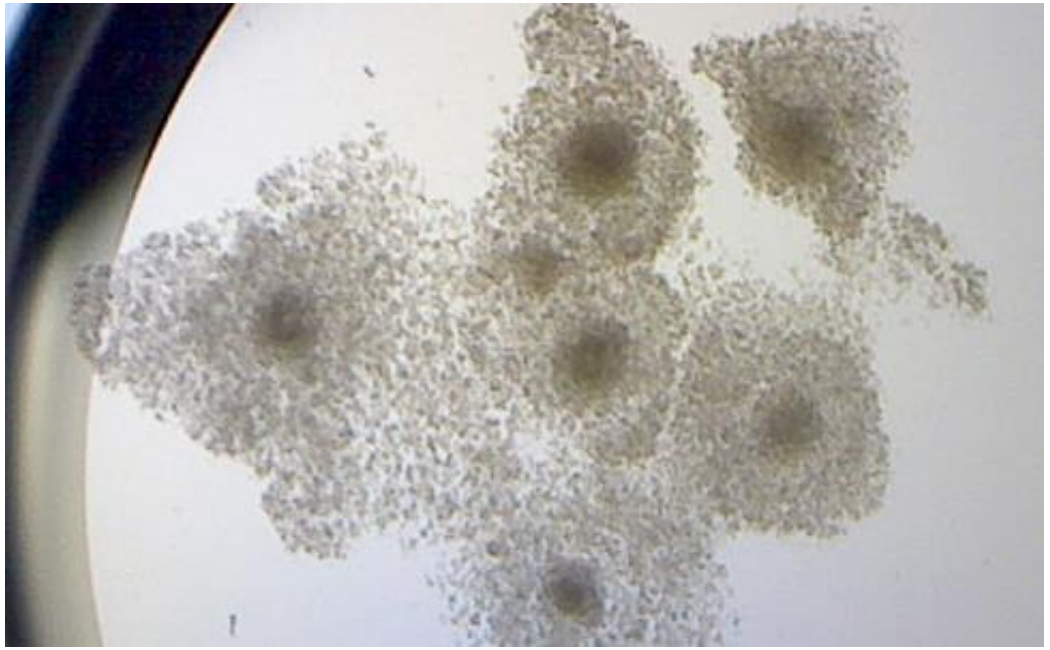
Oocyte Retrieval





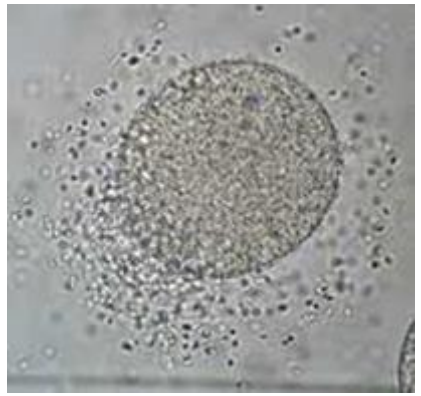
Hunting for Eggs





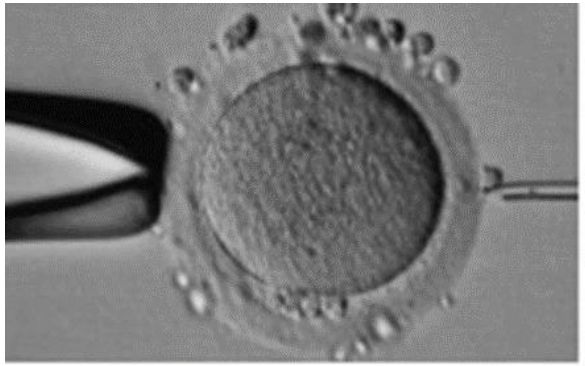
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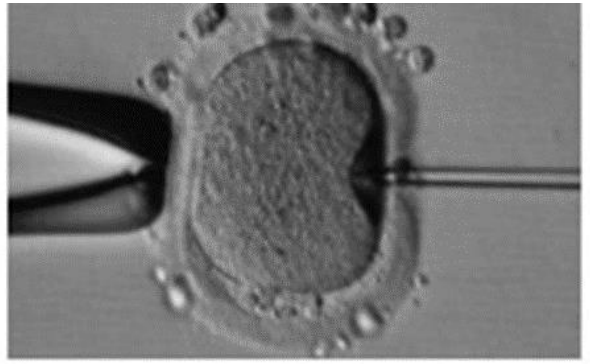
Eggs and Sperm and Intracytoplasmic Sperm Injection (ICSI)

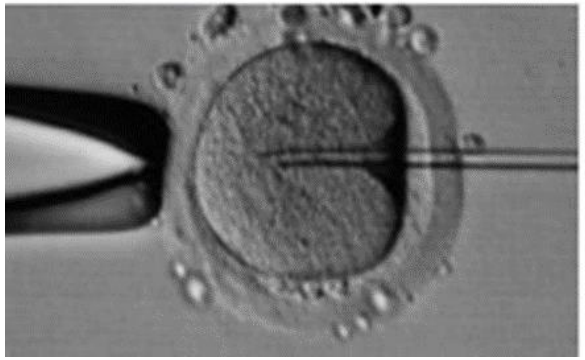


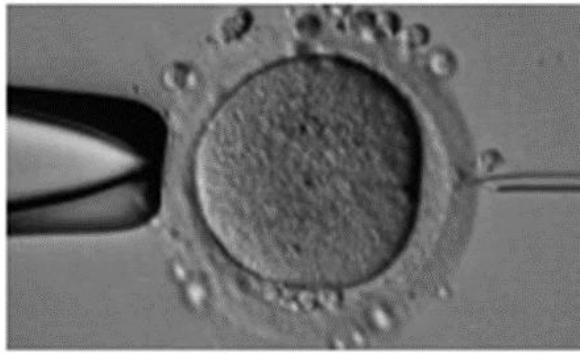




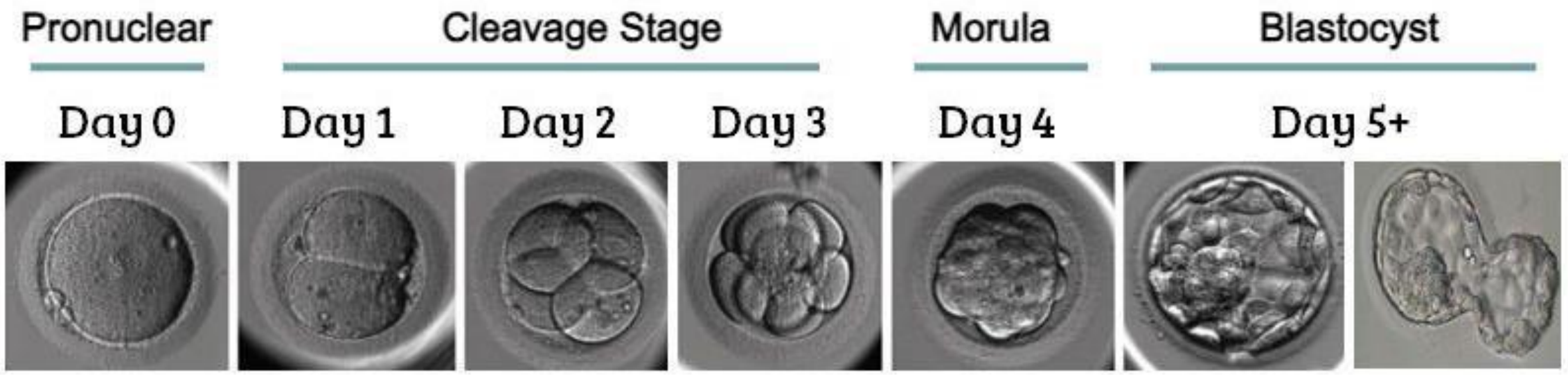








Embryo Development



Pre-Implantation Genetic Testing (PGT)



- PGT-A (Aneuploidy)
- PGT-M (Monogenic)
  - Cystic Fibrosis
  - Fragile X
  - BRCA
- PGT-SR (Structural Rearrangement)
  - Translocations
  - Inversions
- PGT-P (polygenic)

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PGT-A

PGT

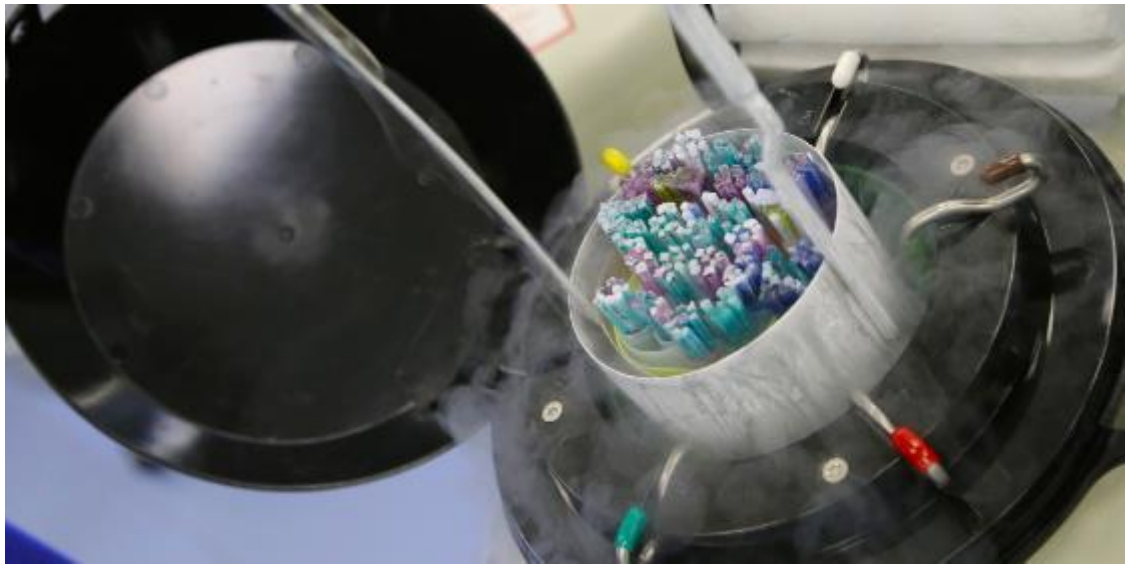
Age of Woman at Retrieval	< 35	35 - 37	38 - 40	41 - 42	> 42
Number of Transfers	22230	9705	5703	2129	1815
Average Number of Embryos Transferred	1.1	1.2	1.3	1.5	1.7
% Cycles with Single Embryo Transfer	**87.9 %**	**84.1 %**	**74.5 %**	**63.0 %**	**58.1 %**
Live Birth Per Single Embryo Transfer	**46.6 %**	**41.0 %**	**34.4 %**	**25.1 %**	**20.6 %**

No PGT

Age of Woman at Retrieval	< 35	35 - 37	38 - 40	41 - 42	> 42
Number of Transfers	38024	26170	20287	6641	3479
Average Number of Embryos Transferred	1.0	1.0	1.0	1.0	1.0
% Cycles with Single Embryo Transfer	**96.8 %**	**97.5 %**	**97.6 %**	**97.8 %**	**96.6 %**
Live Birth Per Single Embryo Transfer	**54.5 %**	**53.3 %**	**52.1 %**	**49.8 %**	**47.4 %**

SART, 2023

Embryo Cryopreservation

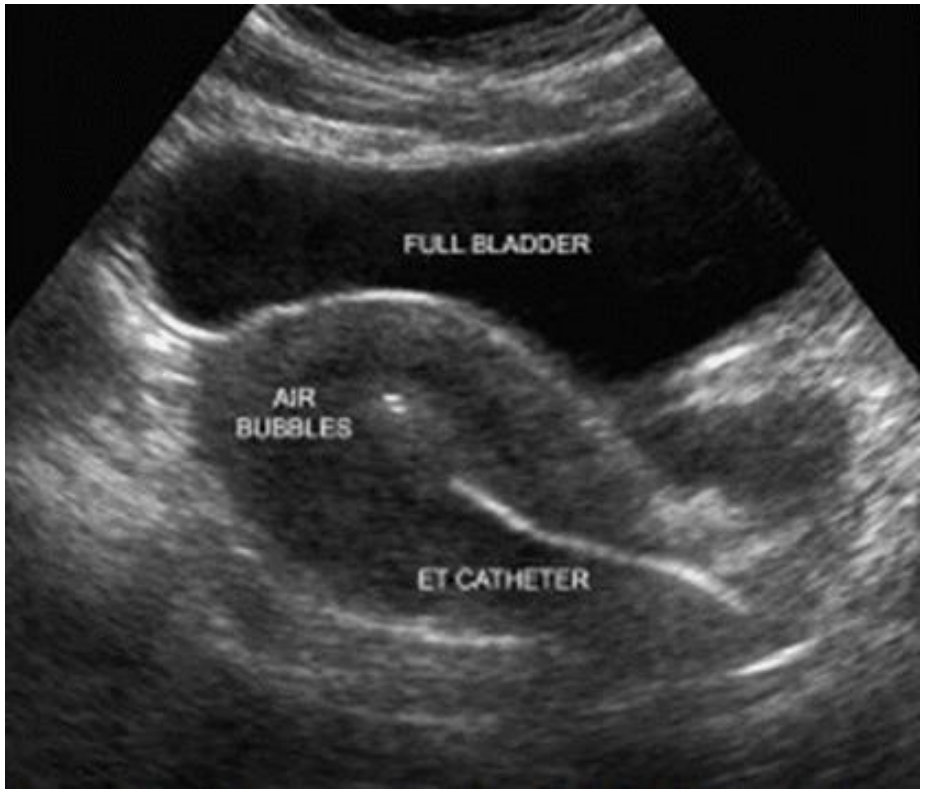




rma Philadelphia

Embryo Transfer





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Success

LIVE BIRTHS PER INTENDED EGG RETRIEVAL (ALL EMBRYO TRANSFERS)

What is the chance of live birth with an intended egg retrieval?

Age of woman	< 35	35 - 37	38 - 40	41 - 42	> 42
Number of cycle starts	54005	37881	37192	19149	14307
Singleton Births	**40.7 %**	**28.9 %**	**17.9 %**	**8.6 %**	**2.7 %**
Live Births	41.7 %	29.6 %	18.5 %	8.8 %	2.8 %

SART, 2023

Endometriosis: Treatment

- GnRH analog (Lupron)
- Progesterone
- Oral contraceptive
- Orlissa
- Surgery



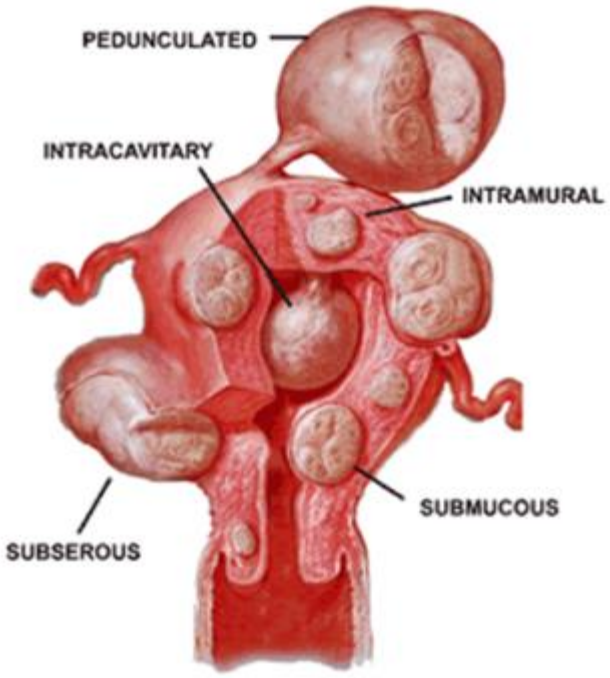
Cochrane Library

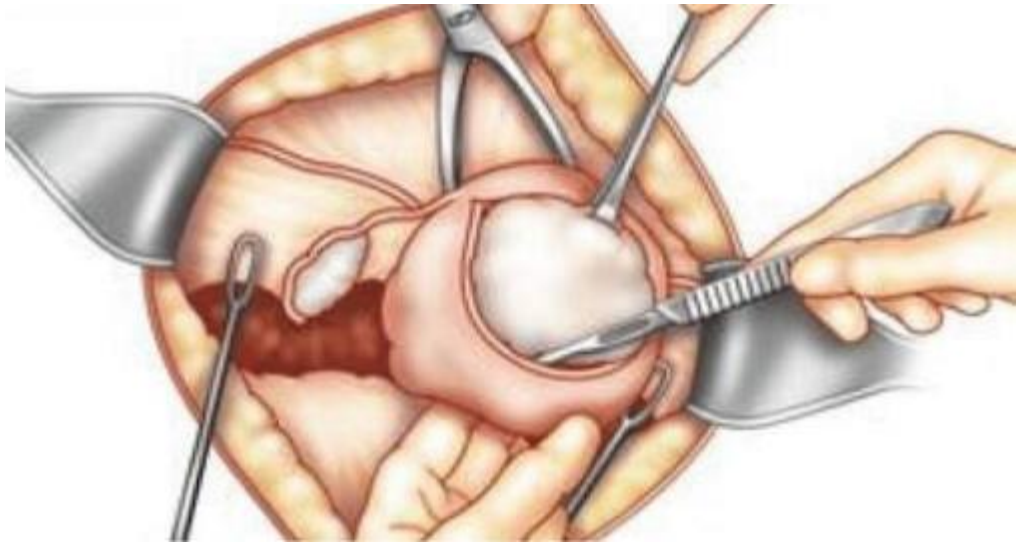
Cochrane Database of Systematic Reviews

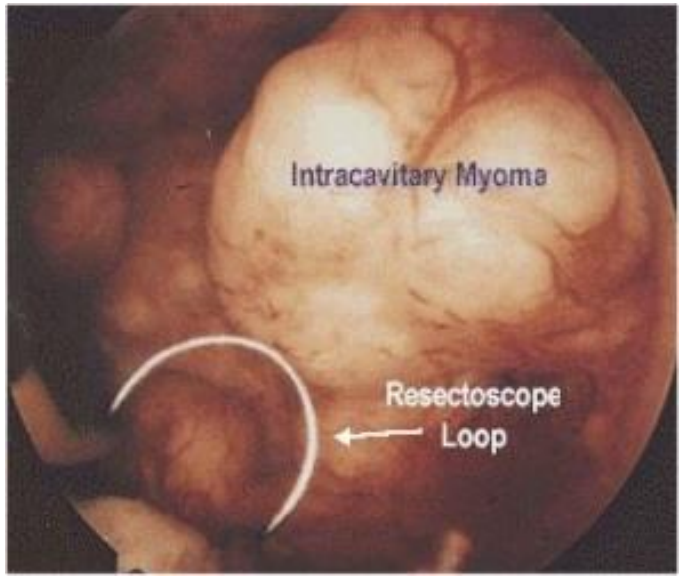
Endometriosis: an overview of Cochrane Reviews (Review)

Brown J, Farquhar C

Uterine Abnormalities: Treatment







Uterine Abnormalities: Treatment

Effect of myomectomy on fertility: intramural fibroids (fibroids in situ controls).

Outcome	Number of studies/ substudies	Relative risk	95% confidence interval	Significance
Clinical pregnancy rate	2	3.765	0.470–30.136	Not significant
Implantation rate	0	—	—	—
Ongoing pregnancy/live birth rate	1	1.671	0.750–3.723	Not significant
Spontaneous abortion rate	1	0.758	0.296–1.943	Not significant
Preterm delivery rate	0	—	—	—

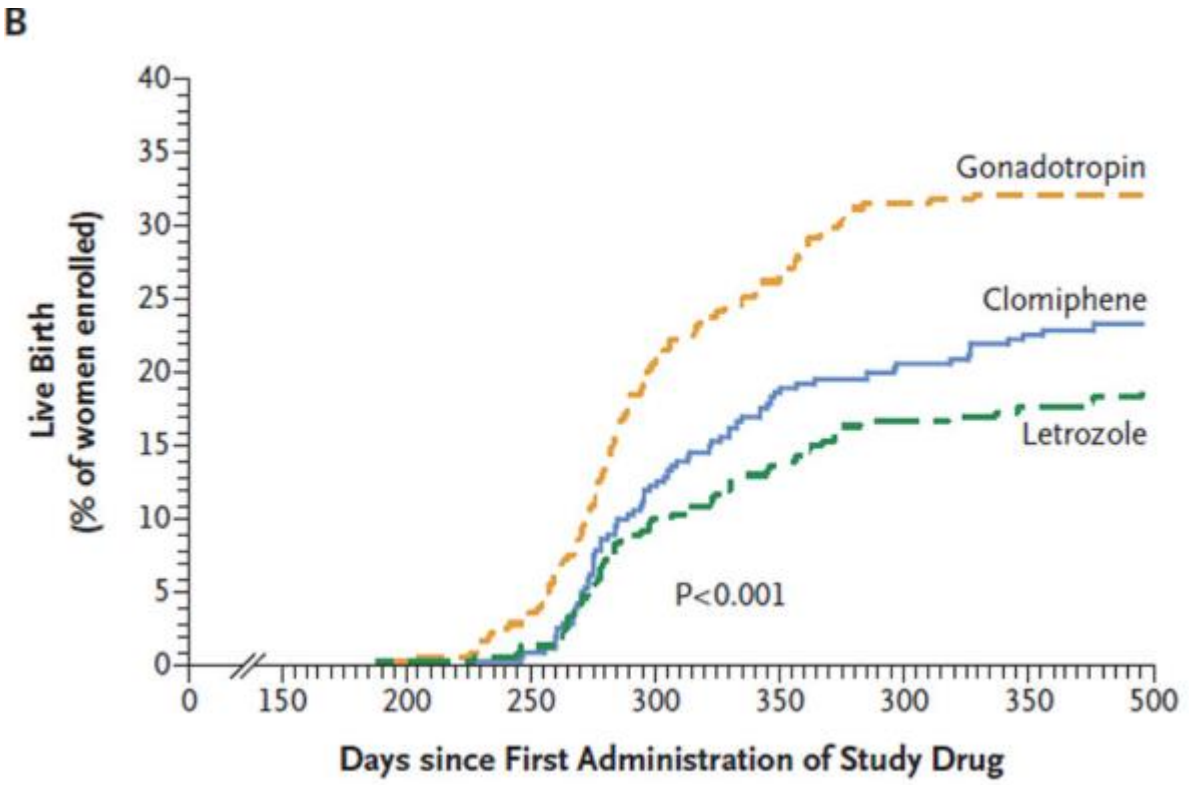
Effect of fibroids on fertility: submucous fibroids.

Outcome	Number of studies/ substudies	Relative risk	95% confidence interval	Significance
Clinical pregnancy rate	4	0.363	0.179–0.737	$P=.005$
Implantation rate	2	0.283	0.123–0.649	$P=.003$
Ongoing pregnancy/live birth rate	2	0.318	0.119–0.850	$P<.001$
Spontaneous abortion rate	2	1.678	1.373–2.051	$P=.022$
Preterm delivery rate	0	—	—	—

Pitts, F&S, 2009

Unexplained Infertility: Treatment



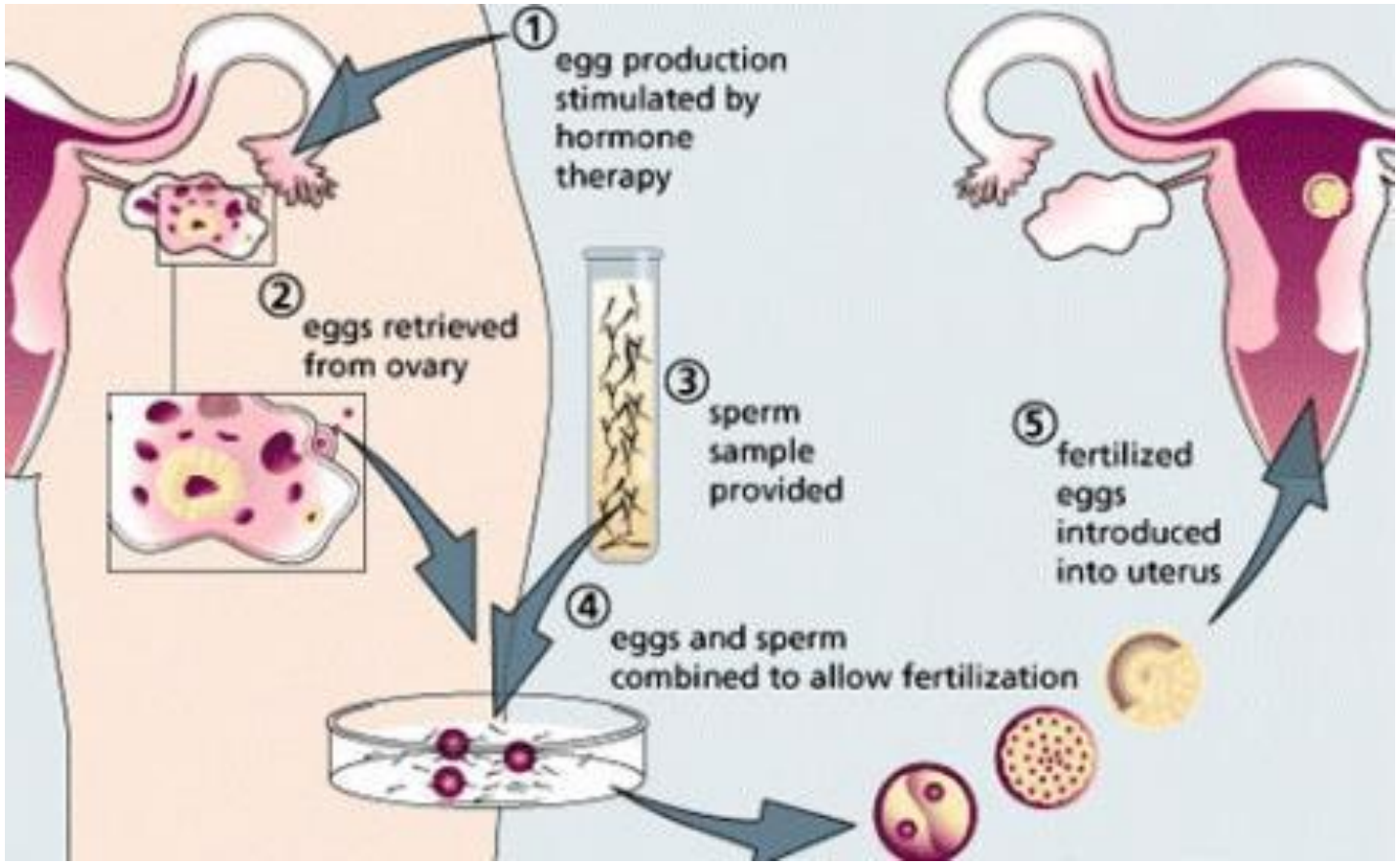


nejm2015

Unexplained Infertility: Treatment



Unexplained Infertility: Treatment





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Egg Freezing

